

31.9 ALPhA-Vision: A Real-Time Always-On Vision Processor with 787 μ s Face Detection Latency in <5mW

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Abstract

ALPhA-Vision is an always-on low-power subsystem for DNN-inference-based vision tasks in edge SoCs. Flexible and programmable, the subsystem supports CNN and ViT inference and employs hardware/software co-design to enable fully end-to-end execution with no

external memory accesses. Fine-grained power management features to mitigate leakage enable the subsystem to perform face detection with 787 μ s latency and 99.3% detection accuracy with 4.6mW average power at 60fps.

Always-on computer vision (CV) tasks are key workloads in applications such as user presence detection, safety monitoring, and gesture-based control, with always-on vision (AoV) poised to become even more prevalent as embodied artificial intelligence enters the mainstream [1]. In consumer devices, use cases such as face detection for opportunistic laptop display sleep can enable significant system-level power savings if a low-latency vision pipeline can be continuously executed in a low power footprint. Typical vision pipelines include both deep neural network (DNN) inference tasks and classical CV kernels, requiring a flexible and capable platform for end-to-end execution. GPU-based edge SoCs can execute these workloads, but their high power consumption (≥ 10 W) limits their utility for AoV applications [2].

This work presents ALPhA-Vision, an Always-on Low-Power Accelerator for vision tasks that can serve as a dedicated subsystem in a larger high-performance SoC, enabling the rest of the SoC to sleep while the standalone subsystem runs continuously in a low power footprint (see Fig. 31.9.1). The programmable subsystem supports a variety of different DNN inference workloads, including both convolutional neural networks (CNNs) and vision transformers (ViTs), and specialized hardware accelerates not only matrix multiplication (GEMM) kernels, but many other layer types commonly found in CV DNNs, ensuring that non-GEMM operations do not bottleneck low-latency execution. Aggressive quantization and a capacity-maximizing hardware implementation enable end-to-end localized processing of image inputs without accessing external memory. Hardware-software codesign for fine-grained power management enables a sub-5mW power budget even at high framerates. A companion full-stack compiler flow lowers application source code to execution binaries after performing hardware-compliant quantization-aware training (QAT) to maintain task accuracy. Together, these components enable end-to-end execution of AoV pipelines without waking the high-power CPU, GPU, or main memory system of the full SoC.

Figure 31.9.2 shows the ALPhA-Vision architecture, which comprises a deep-learning accelerator (DLA) to execute GEMMs, a near-memory processor (NMP) dedicated to low-compute-intensity kernels, a 32b RISC-V Rocket processor (RV) [3], and a global memory system containing 2.125MB of SRAM, which fills much of the available chip area to accommodate fully resident weights and activations. Unlike prior works that target ultra-low-power edge devices using specialized low-leakage processes [4-10], ALPhA-Vision is designed as a subsystem for a larger SoC and therefore uses a standard high-performance process. Accordingly, SRAM leakage is a large component of overall power consumption, and minimizing this leakage power was a first-class consideration for system design. The most important factor in reducing static power consumption is for the system to “race to sleep” so it can spend as much time as possible in a low-power sleep mode. To accomplish this, compute units are optimized to target sub-ms end-to-end latency for typical workloads so that the system can sleep for most of each frame, even in high-framerate (60fps) scenarios.

The ALPhA-Vision DLA accelerates convolution and GEMM kernels that make up the bulk of DNN compute. The DLA, which supports both INT8 and INT4 with microscaling (INT8 scale factors with a block size of 32) [11,12] math operations, performs matrix-vector multiplication in 32 parallel lanes of 32-element vector MACs (16 for INT8) each cycle, for a total of 1024 (512 for INT8) MACs/cycle when fully utilized. Weights and activations are initially stored in global memory, with local buffers, collectors, and address generators orchestrating an efficient output-stationary, local-weight-stationary dataflow to maximize reuse [13], while affording high utilization on a range of different GEMM shapes. A post-processing unit (PPU) performs bias addition, scaling, and simple activation operations, such as ReLU and Tanh.

The compute acceleration afforded by the DLA is not sufficient to realize low latency on end-to-end DNN inference workloads, as a “long tail” of operations with low compute intensity would dominate the remaining runtime. The NMP, a programmable vector engine with high (128B/cycle) bandwidth to global memory, accelerates these remaining workloads. The NMP supports depthwise convolutions (DWConv) with a specialized engine performing 32 MACs/cycle, non-GEMM operations such as Groupnorm and SoftMax, elementwise operations, and memory-manipulation operations like shuffle and transpose, all of which are performed in INT8. A dedicated perspective warp engine offloads this common classical CV function from the CPU.

The global memories were co-designed with the accelerators to avoid memory-induced bottlenecks, while maximizing capacity. The memory system consists of a global weight memory (GWM, 16 banks, 68B/cycle peak read bandwidth) that stores most weights and a global scratchpad (GS, 64 banks, 136B/cycle peak read bandwidth, 65B/cycle peak write bandwidth) that stores the input image, intermediate activations, DWConv layer weights, and the RV program. The high GS bank count enables fine-grained memory management and supports multiple parallel accesses to different banks, with 14 physical banks statically grouped into virtual bank clusters for higher bandwidth. Virtual-to-physical address translation is hardware managed by accelerator address generators at runtime and can be reconfigured per DNN layer. The global memories minimize large, expensive crossbar structures by a compiler-enforced requirement that each bank be reserved for access by at most one requestor at a time.

The ALPhA-Vision design minimizes static power, the dominant component of energy per frame, both through direct optimization and by minimizing active-mode latency to maximize the time the subsystem is in sleep mode (see Fig. 31.9.3). Each of the GWM and GS banks can be independently placed into retention (power-gated periphery) or sleep (power-gated macros) modes, dramatically reducing leakage power, and these settings can be adjusted on-the-fly for fine-grained sleep state control during active-mode execution [14]. Outside of clock trees, HVT standard cells are favored to minimize leakage. In addition to hardware-managed leaf-level clock gating (CG), the system implements software-controlled block-level CG for the DLA and NMP, as well as a software-programmable root CG that halts the entire clock distribution network at its root, leaving only a single 31b counter active as a timer to restart the clock. To reduce control overhead and the associated latency, the DLA and NMP have deep configuration memories enabling up to 384 kernels to be fused together and executed back-to-back with no intervention from the RV. To further reduce control overhead and to minimize the latency of any workloads that lack hardware acceleration support, the RV supports a double-clock execution mode in which its clock runs at twice the frequency of the rest of the system with zero-cycle clock domain crossing latency between the 2 $\times$ and 1 $\times$ domains.

Figure 31.9.4 provides an overview of the application software flow used to execute end-to-end DNN inference workloads on ALPhA-Vision. Application software in PyTorch is quantized with a per-layer configuration, which accounts for both hardware constraints and maintaining accuracy, that is used to perform QAT, yielding quantized per-layer weights, biases, and scaling factors that can be executed in hardware. The software stack also allocates memory in the GWM and GS, tracking the liveness of activation data to maximize the number of activation memory banks that can be placed in retention or sleep mode during each layer execution while striding data across banks to accommodate high-bandwidth accesses. Timeloop [15] uses a high-level hardware model to map each GEMM onto the DLA, selecting loop nest dimensions to maximize utilization. Finally, the quantized layers are fused into one or more execution kernels. Using this flow, the weight and peak activation memory footprints of Yolo5n-0.5 [16] are reduced to 318KB and 461KB, respectively, while maintaining task accuracy, enabling fully on-chip execution with no external memory accesses.

A standalone testchip prototype of the ALPhA-Vision subsystem was taped out in 16nm with an active area of 4.20mm². In contrast to prior works that run at very low frequencies [4,7,17,18], the testchip operates at 359MHz at 0.6V, a moderate clock period target that minimized the need for LVT cell swaps and gate upsizing, while minimizing latency for race-to-sleep. Weights and input activations are loaded into the scratchpads before each measurement test, and the RV controller programs SRAM sleep states and block CG between kernel launches, as well as executing the non-maximum suppression (NMS) workload at the end of Yolo5.

Figure 31.9.5 summarizes key measurement results, which are measured at 0.6V and 359MHz (RV at 718MHz) unless otherwise noted. We evaluate the end-to-end performance of the system with face detection using Yolo5 trained on WiderFace [19] and tested on 192 $\times$ 192 grayscale input images. At 0.6V, the system achieves a face detection latency of 787ms and a peak throughput of 1270fps, but a more realistic use case operates at 60fps with 95.3% of each frame spent in sleep mode. The runtime breakdown demonstrates that the accelerators are effective at mitigating compute bottlenecks, with non-GEMM operations

using just 21% of active cycles. At 60fps, the system operates with 4.6mW average power, compared to 1.8mW of unavoidable leakage power with clocks halted and all SRAMs in sleep mode. Our power-saving techniques reduce active and sleep energy by 21% and 94% respectively, with 10% active energy savings from partitioning the end-to-end workload into multiple kernels to allow fine-grained reprogramming of SRAM sleep states. To showcase the versatility of the subsystem, we also evaluate performance on a wide range of kernels from MobileVITv2 [20] and YOLOv5, as well as a perspective warp kernel. Across these diverse workloads, the DLA achieves an average compute utilization of 81% and the NMP achieves an average bandwidth utilization of 86%. Figure 31.9.6 shows the comparison to prior works, and Fig. 31.9.7 shows the die photo and kernel measurement results. Taken together, the measurement results demonstrate that the ALPhA-Vision subsystem can enable AoV applications in power-constrained edge devices.

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References:

- [1] J. Duan et al., "A Survey of Embodied AI: From Simulators to Research Tasks," *IEEE TETCI*, vol. 6, no. 2, pp. 230–244, 2022. <https://doi.org/10.1109/TETCI.2022.3141105>
- [2] NVIDIA, *Jetson Orin NX Module*. <https://developer.nvidia.com/downloads/jetson-orin-x-series-data-sheet>
- [3] K. Asanović et al., "The Rocket Chip Generator," Tech. Rep., EECS Department, University of California, Berkeley, 2016. <https://www2.eecs.berkeley.edu/Pubs/TechRpts/2016/EECS-2016-17.html>
- [4] Z. Fan et al., "Audio and Image Cross-Modal Intelligence via a 10 TOPS/W 22 nm SoC with Back-Propagation and Dynamic Power Gating," *IEEE Symp. VLSI Circuits*, pp. 18–19, 2022. <https://doi.org/10.1109/VLSITechnologyandCirc46769.2022.9830226>
- [5] V. Jain et al., "TinyVers: A 0.8–17 TOPS/W, 1.7 μ W–20 mW, Tiny Versatile System-on-Chip with State-Retentive eMRAM for Machine Learning Inference at the Extreme Edge," *IEEE Symp. VLSI Circuits*, pp. 20–21, Jun. 2022. <https://doi.org/10.1109/VLSITechnologyandCirc46769.2022.9830409>
- [6] P. Jokic et al., "A Sub-mW Dual-Engine ML Inference System-on-Chip for Complete End-to-End Face-Analysis at the Edge," *IEEE Symp. VLSI Circuits*, 2021. <https://doi.org/10.23919/VLSICircuits52068.2021.9492401>
- [7] H. An et al., "A 170 μ W Image Signal Processor Enabling Hierarchical Image Recognition for Intelligence at the Edge," *IEEE Symp. VLSI Circuits*, 2020. <https://doi.org/10.1109/VLSICircuits18222.2020.9162810>
- [8] I. Miro-Panades et al., "Samurai: a 1.7MOPS–36GOPS Adaptive Versatile IoT Node with 15,000 $\times$ Peak-to-Idle Power Reduction, 207 ns Wake-Up Time and 1.3 TOPS/W ML Efficiency," *IEEE Symp. VLSI Circuits*, 2020. <https://doi.org/10.1109/VLSICircuits18222.2020.9163000>
- [9] D. Garrett et al., "A 1mW Always-on Computer Vision Deep Learning Neural Decision Processor," *ISSCC*, pp. 158–159, 2023. <https://doi.org/10.1109/ISSCC42615.2023.10067588>
- [10] F. Conti et al., "A 12.4TOPS/W @ 136 GOPS AI-IoT System-on-Chip with 16 RISC-V, 2-to-8b Precision-Scalable DNN Acceleration and 30%-Boost Adaptive Body Biasing," *ISSCC*, pp. 21–23, 2023. <https://doi.org/10.1109/ISSCC42615.2023.10067643>

- [11] S. Dai et al., "VS-Quant: Per-vector Scaled Quantization for Accurate Low-Precision Neural Network Inference," *MLSys*, pp. 873–884, 2021. <https://doi.org/10.48550/arXiv.2102.04503>
- [12] B. D. Rouhani et al., "Microscaling Data Formats for Deep Learning," *arXiv preprint*, arXiv:2310.10537, 2023. <https://doi.org/10.48550/arXiv.2310.10537>
- [13] B. Keller et al., "A 95.6-TOPS/W Deep Learning Inference Accelerator With Per-Vector Scaled 4-bit Quantization in 5 nm," *IEEE JSSC*, vol. 58, no. 4, pp. 1129–1141, 2023. <https://doi.org/10.1109/JSSC.2023.3234893>
- [14] K. Prabhu et al., "MINOTAUR: A Posit-Based 0.42–0.50-TOPS/W Edge Transformer Inference and Training Accelerator," *IEEE JSSC*, vol. 60, no. 4, pp. 1311–1323, 2025. <https://doi.org/10.1109/JSSC.2025.3545731>
- [15] A. Parashar et al., "Timeloop: A Systematic Approach to DNN Accelerator Evaluation," *IEEE ISPASS*, pp. 304–315, 2019. <https://doi.org/10.1109/ISPASS.2019.00042>
- [16] Q. DeLong et al., "YOLO5Face: Why Reinventing a Face Detector," *arXiv preprint*, arXiv:2105.12931, 2021. <https://doi.org/10.48550/arXiv.2105.12931>
- [17] Z. Yuan et al., "A 65nm 24.7 μ J/Frame 12.3mW Activation-Similarity-Aware Convolutional Neural Network Video Processor Using Hybrid Precision, Inter-Frame Data Reuse and Mixed-Bit-Width Difference-Frame Data Codec," *ISSCC*, pp. 232–234, 2020. <https://doi.org/10.1109/ISSCC19947.2020.9063155>
- [18] S. Kwon et al., "Monolithic in-Memory Computing Microprocessor for End-to-End DNN Inference in MRAM-Embedded 28 nm CMOS Technology with 1.1 Mb Weight Storage," *ISSCC*, pp. 610–612, 2025. <https://doi.org/10.1109/ISSCC49661.2025.10904609>
- [19] S. Yang et al., "WIDER FACE: A Face Detection Benchmark," *CVPR*, pp. 5525–5533, 2016. <https://doi.org/10.1109/CVPR.2016.596>
- [20] S. Mehta and M. Rastegari, "Separable Self-attention for Mobile Vision Transformers," *TMLR*, 2023. <https://doi.org/10.48550/arXiv.2206.02680>
- [21] G. Fanelli et al., "Random Forests for Real Time 3D Face Analysis," *ICCV*, vol. 101, no. 3, pp. 437–458, 2013. <https://doi.org/10.1007/s11263-012-0549-0>
- [22] V. Jain and E. Learned-Miller, "FDDB: A Benchmark for Face Detection in Unconstrained Settings," UMass Amherst Tech. Rep. UM-CS-2010-009. <https://people.cs.umass.edu/~elm/papers/fddb.pdf>
- [23] A. Gupta et al., "CogniVision: End-to-End SoC for Always-on Smart Vision with mW Power in 40 nm," *IEEE Symp. VLSI Circuits*, 2024. <https://doi.org/10.1109/VLSITechnologyandCirc46783.2024.10631426>
- [24] E. Chang et al., "A 12-nm 0.62–1.61 mW Ultra-Low Power Digital CIM- based Deep-Learning System for End-to-End Always-on Vision," *IEEE Symp. VLSI Circuits*, 2023. <https://doi.org/10.23919/VLSITechnologyandCirc57934.2023.10185296>

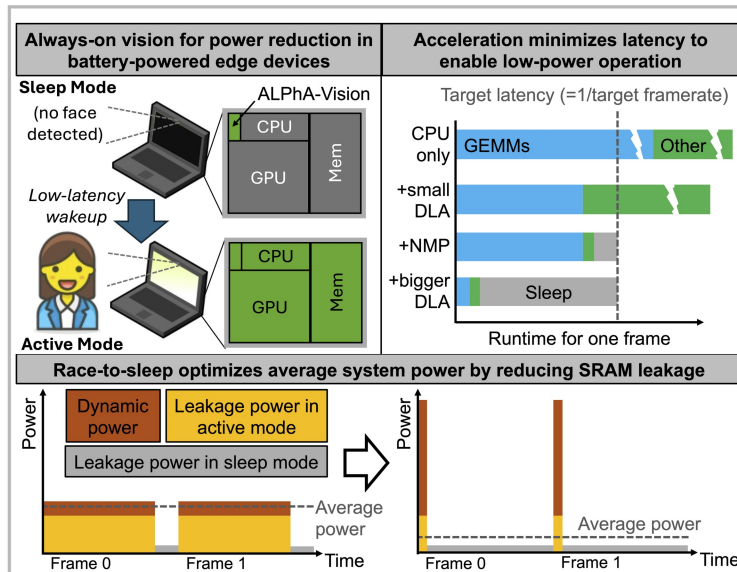


Figure 31.9.1: Motivation and high-level design principles for the ALPhA-Vision subsystem.

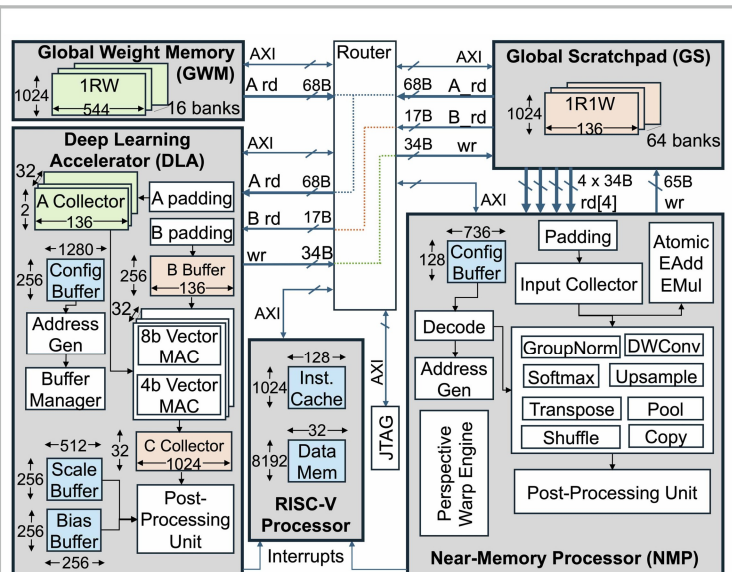


Figure 31.9.2: Block diagram of the ALPhA-Vision architecture.

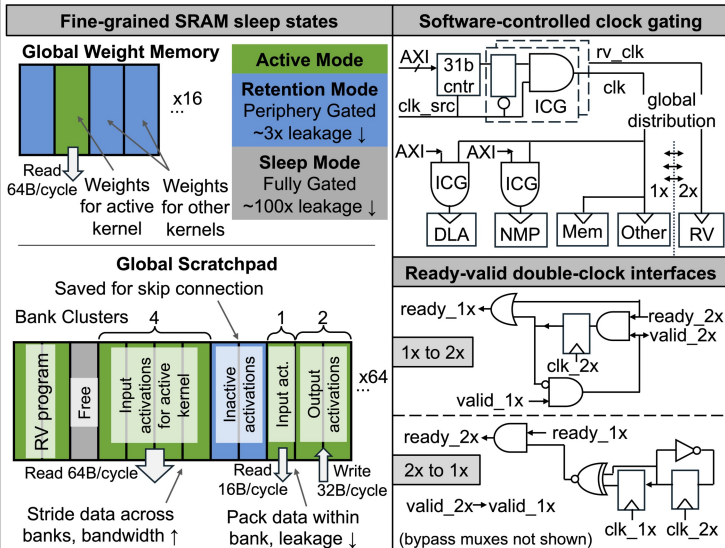


Figure 31.9.3: Overview of power-saving features in the ALPhA-Vision subsystem.

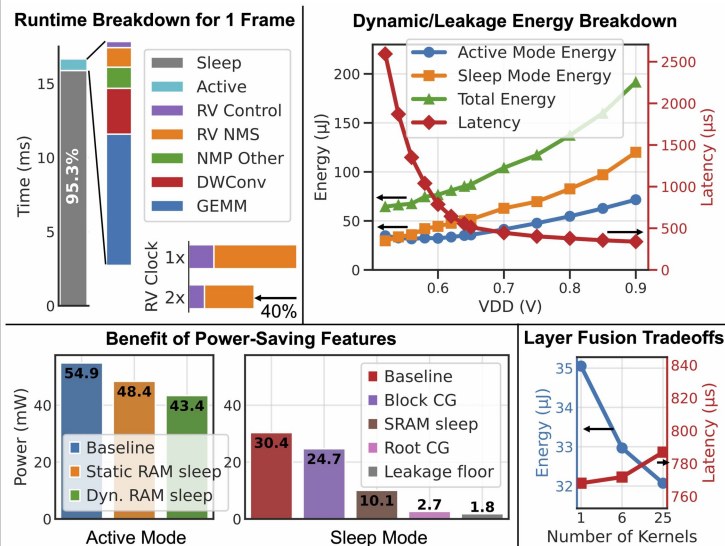


Figure 31.9.5: ALPhA-Vision measurement results on end-to-end Yolov5 face detection at 60fps.

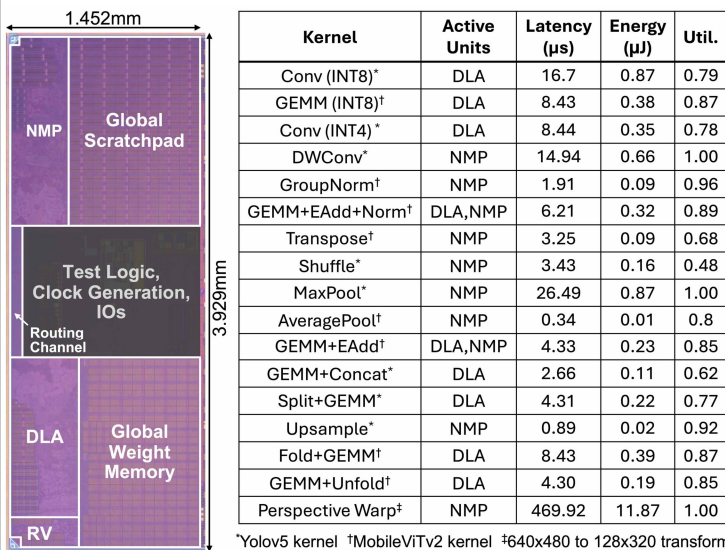


Figure 31.9.7: Annotated die plot and evaluation summary of DNN inference kernels (50% weight/activation sparsity).

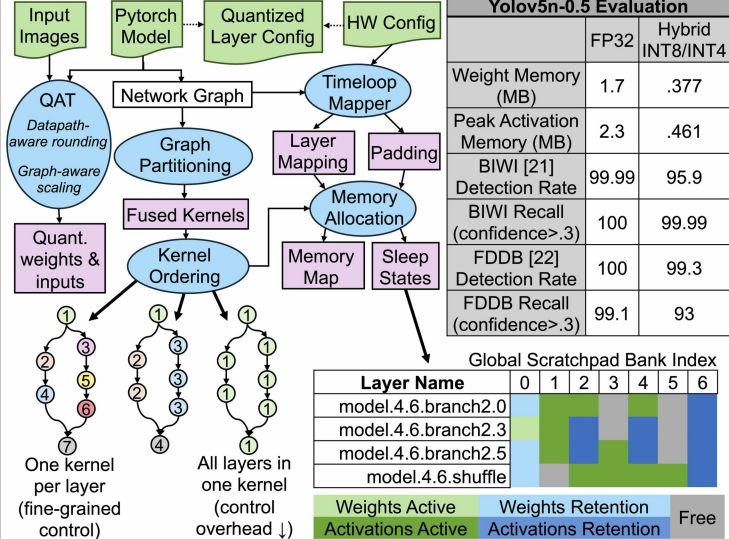


Figure 31.9.4: The ALPhA-Vision application software flow and impact of quantization of YoloV5 face detection.

	This Work	[23]	[24]	[18]	[6]
Application	Always-on vision	Always-on vision	Always-on vision	Low-power inference	Face analysis
Process (nm)	16	40	12	28	22
Area (mm²)	4.20	30	1.76	19.95	3.4
V_{DD} (V)	0.52-0.9	0.6-1.1	0.6	1	0.59-0.72
Frequency (MHz)	90-830	350	100-200	62.5	180-220
Power (mW)	3.9-11.5 ¹	2.1 ⁵	0.62-1.61 ⁷	19-32 ⁶	0.16-12.77 ⁹
Memory (MB)	2.39	1.798	0.5	0.125	1.176
Throughput (GOPS)	1465 ²	2250	51	129	5
Energy Eff. (TOPS/W)	16.3 ³ (4b)	29 ⁶ (8b)	577.8 ⁸ (8b/4b)	20 ⁶ (1b)	1.1 ⁹ (1b/16b)
Latency (ms)	0.787 ^{1,4}	33 ⁵	67-500 ⁷	-	50-1000 ⁹
Energy/frame (μJ)	76.7 ^{1,4}	91 ⁵	107-310 ⁷	-	409 ⁹
Energy/pixel (nJ)	2.1 ^{1,4}	1.16 ⁵	6.5-18.9 ⁷	-	4.0 ⁹
FDDb Face Detection Accuracy	99.3%	-	-	91.3%	98%
Fully On-Chip End-to-End Execution	Yes	Yes	No	No	Yes

One MAC = 2 ops, other ops ignored. ¹Yolov5 face detection. ²0.9V, 830MHz. ³Benchmark GEMM kernel, 50% sparse, 0.54V, 146MHz. ⁴0.6V, 359MHz. ⁵SqueezeNet image classification. ⁶Workload unspecified. ⁷MobileNet human detection. ⁸DLA only. ⁹BDT face detection + CNN face recognition.

Figure 31.9.6: Comparison table. All measurements for this work report full system power, including leakage power.